
Notes for ‘Integration by substitution’

Important Ideas and Useful Facts:

- (i) The word *integration* has two meanings: The word *integration* can refer to the process of forming an antiderivative of a function $f(x)$, expressed as the indefinite integral

$$\int f(x) \, dx .$$

The word *integration* can also refer to the process of finding or evaluating the area under the curve $y = f(x)$ over an interval $[a, b]$, represented by the definite integral

$$\int_a^b f(x) \, dx .$$

The terminology always has to be read in context. Of course, the two meanings are closely related, because if $F(x)$ is any antiderivative of $f(x)$ then the Fundamental Theorem of Calculus tells us that

$$\int_a^b f(x) \, dx = F(b) - F(a) .$$

- (ii) **Integration by substitution:** The method of *integration by substitution* typically refers to a method of antidifferentiation, and says that

$$\int \left(f(u) \frac{du}{dx} \right) dx = \int f(u) \, du ,$$

or, equivalently, in function notation, where $u = g(x)$ for some differentiable function g ,

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du .$$

The first form, expressed in Leibniz notation, is particularly easy to remember, because one can think of the differential dx , appearing at the end of the expression on the left-hand side, as cancelling with the differential dx appearing in $\frac{du}{dx}$, as if $\frac{du}{dx}$ were an ordinary fraction involving numbers. It is common to move freely backwards and forwards between the left and right-hand sides using equations involving differentials.

The method can also be applied using definite integrals, where we add appropriate terminals, a and b for the variable x , on the left-hand side, and $g(a)$ and $g(b)$ for the variable u , on the right-hand side:

$$\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du .$$

Examples and derivations:

1. The rule for integration by substitution follows from the Chain Rule for differentiation. To see this, suppose that y is an antiderivative of $f(u)$ with respect to the variable u , so its derivative is $f(u)$, that is,

$$\frac{dy}{du} = f(u) .$$

We can express this relationship using an equation involving an indefinite integral:

$$\int f(u) du = y + C , \quad (1)$$

for some constant of integration C . But, by the Chain Rule,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = f(u) \frac{du}{dx} ,$$

so that y is also an antiderivative of $f(u) \frac{du}{dx}$, but now with respect to x . Again, this can be expressed as an equation involving an indefinite integral:

$$\int \left(f(u) \frac{du}{dx} \right) dx = y + C , \quad (2)$$

for some constant of integration C , which may be taken to be the same constant appearing earlier. Hence, by (1) and (2), we have

$$\int \left(f(u) \frac{du}{dx} \right) dx = y + C = \int f(u) du ,$$

which finally gives us the integration by substitution formula.

2. Find the indefinite integral $\int 6x(x^2 + 1)^2 dx$ by making a substitution $u = x^2 + 1$.

Solution: Put $u = x^2 + 1$, so that $\frac{du}{dx} = 2x$, which we may write as $du = 2x dx$. Putting everything in terms of u , by reorganising the integrand and constants, we get

$$\int 6x(x^2 + 1)^2 dx = \int 3(x^2 + 1)^2 (2x dx) = \int 3u^2 du = u^3 + C = (x^2 + 1)^3 + C .$$

3. Find the indefinite integral $\int e^{-x} dx$.

Solution: Put $u = -x$. Hence $\frac{du}{dx} = -1$, so that $du = -dx$, which can be written as $dx = -du$. Expressing everything in terms of u , including the differential, we have

$$\int e^{-x} dx = \int e^u (-du) = - \int e^u du = -e^u + C = -e^{-x} + C .$$

4. Find the definite integral $\int_0^{\ln 2} e^{-x} dx$.

Solution: Using the indefinite integral from the previous exercise, we have

$$\int_0^{\ln 2} e^{-x} dx = [-e^{-x}]_0^{\ln 2} = -e^{-\ln 2} - (-e^{-0}) = -(e^{\ln 2})^{-1} - (-1) = -2^{-1} + 1 = \frac{1}{2} .$$

5. Find the indefinite integral $\int \left(\frac{x-1}{2}\right)^3 dx$.

Solution: Put $u = \frac{x-1}{2}$. Then $\frac{du}{dx} = \frac{1}{2}$, so that $du = \frac{1}{2}dx$, which can be rewritten as $dx = 2du$. Expressing everything in terms of u , including the differential, we have

$$\int \left(\frac{x-1}{2}\right)^3 dx = \int u^3 (2du) = 2 \int u^3 du = \frac{2u^4}{4} + C = \frac{u^4}{2} + C = \frac{(x-1)^4}{32} + C.$$

6. Find the definite integral $\int_1^2 \left(\frac{x-1}{2}\right)^3 dx$.

Solution: Using the indefinite integral from the previous exercise, we have

$$\int_1^2 \left(\frac{x-1}{2}\right)^3 dx = \left[\frac{(x-1)^4}{32}\right]_1^2 = \frac{1}{32}[(x-1)^4]_1^2 = \frac{1}{32}((2-1)^4 - (1-1)^4) = \frac{1}{32}.$$

7. Find the definite integral $\int_{\pi/4}^{\pi/2} \sin(2x) dx$.

Solution: Put $u = 2x$, so that $du = 2dx$, giving $dx = \frac{1}{2}du$. Putting everything in terms of u , we get the following indefinite integral:

$$\int \sin(2x) dx = \int \sin u \left(\frac{1}{2}du\right) = \frac{1}{2} \int \sin u du = \frac{-\cos u}{2} + C = \frac{-\cos(2x)}{2} + C.$$

Now we find the definite integral (keeping the terminals in terms of x):

$$\int_{\pi/4}^{\pi/2} \sin(2x) dx = \frac{1}{2}[-\cos(2x)]_{\pi/4}^{\pi/2} = \frac{1}{2}\left(-\cos(\pi) - \left(-\cos\left(\frac{\pi}{2}\right)\right)\right) = \frac{1}{2}(-(-1)+0) = \frac{1}{2}.$$

8. Find the definite integral $\int_0^{\pi/4} \cos\left(2x - \frac{\pi}{2}\right) dx$.

Solution: Put $u = 2x - \frac{\pi}{2}$, so that $du = 2dx$, giving $dx = \frac{1}{2}du$. Further, when $x = 0$ we have $u = -\frac{\pi}{2}$, and when $x = \pi/4$ we have $u = 0$. Hence, putting everything in terms of u , including converting the terminals, we get

$$\begin{aligned} \int_0^{\pi/4} \cos\left(2x - \frac{\pi}{2}\right) dx &= \int_{-\pi/2}^0 \cos u \left(\frac{1}{2}du\right) = \frac{1}{2} \int_{-\pi/2}^0 \cos u du = \frac{1}{2}[\sin u]_{-\pi/2}^0 \\ &= \frac{1}{2}\left(\sin 0 - \sin\left(-\frac{\pi}{2}\right)\right) = \frac{1}{2}(0 - (-1)) = \frac{1}{2}. \end{aligned}$$

9. Find the indefinite integral $\int xe^{-x^2} dx$.

Solution: Put $u = -x^2$, so that $\frac{du}{dx} = -2x$, so that $du = -2xdx$. Putting everything in terms of u , we get

$$\int xe^{-x^2} dx = -\frac{1}{2} \int e^{-x^2} (-2xdx) = -\frac{1}{2} \int e^u du = -\frac{e^u}{2} + C = -\frac{e^{-x^2}}{2} + C.$$

10. Find the indefinite integral $\int \sin^3 x \cos x \, dx$.

Solution: Put $u = \sin x$, so that $\frac{du}{dx} = \cos x$, so that $du = \cos x dx$. Putting everything in terms of u , we get

$$\int \sin^3 x \cos x \, dx = \int u^3 \, du = \frac{u^4}{4} + C = \frac{\sin^4 x}{4} + C.$$

11. Find the indefinite integral $\int \frac{4x-3}{x^2+1} \, dx$.

Solution: We split the integral into two pieces:

$$\int \frac{4x-3}{x^2+1} \, dx = \int \frac{4x}{x^2+1} \, dx - 3 \int \frac{dx}{x^2+1}.$$

For the first piece, put $u = x^2 + 1$, so that $\frac{du}{dx} = 2x$, so that $du = 2x dx$. Hence

$$\int \frac{4x}{x^2+1} \, dx = 2 \int \frac{1}{x^2+1} (2x dx) = 2 \int \frac{1}{u} \, du = 2 \ln |u| + C = 2 \ln(x^2+1) + C,$$

noting that $x^2 + 1$ is always positive.

Continuing, we have

$$\int \frac{4x-3}{x^2+1} \, dx = \int \frac{4x}{x^2+1} \, dx - 3 \int \frac{dx}{x^2+1} = 2 \ln(x^2+1) - 3 \tan^{-1} x + C.$$

12. Find the definite integral $\int_{-1}^0 x\sqrt{x+1} \, dx$.

Solution: Put $u = \sqrt{x+1} = (x+1)^{1/2}$. Hence

$$\frac{du}{dx} = \frac{1}{2}(x+1)^{-1/2} = \frac{1}{2\sqrt{x+1}} = \frac{1}{2u}.$$

This becomes $du = \frac{dx}{2u}$, which we may rewrite as $dx = 2u du$. Note also that $u^2 = x+1$, so that $x = u^2 - 1$.

Hence, expressing everything in terms of u , we get

$$\begin{aligned} \int x\sqrt{x+1} \, dx &= \int (u^2 - 1)(u)(2u du) = 2 \int (u^2 - 1)u^2 \, du = 2 \int (u^4 - u^2) \, du \\ &= 2 \left(\frac{u^5}{5} - \frac{u^3}{3} \right) + C = \frac{2(x+1)^{5/2}}{5} - \frac{2(x+1)^{3/2}}{3} + C. \end{aligned}$$

We can now evaluate the definite integral:

$$\int_{-1}^0 x\sqrt{x+1} \, dx = \left[\frac{2(x+1)^{5/2}}{5} - \frac{2(x+1)^{3/2}}{3} \right]_{-1}^0 = \frac{2}{5} - \frac{2}{3} = -\frac{4}{15}.$$